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MESAlab: a Pipeline for Mapping the Blue Loop with MESA runs

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Software

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Summary

Modules for Experiments in Stellar Astrophysics (MESA; Paxton et al. (2011), Paxton et al. (2013), Paxton et al. (2015), Paxton et al. (2018), Paxton et al. (2019); Jermyn et al. (2023)) is a widely used open-source software for modelling stellar evolution. In many studies, the computational grids covers thousands of models (e.g., Joyce et al. (2024), which requires a tremendous amount of time and computational effort to process.

To streamline data analysis, the `mesalab` package was developed. This Python-based pipeline is designed to simplify the post-processing of MESA outputs by automatically identifying various stellar evolutionary phases, with a specific focus on the "blue loop"—a blue-ward excursion in the Hertzsprung-Russell Diagram (HRD) for intermediate-mass stars often associated with peculiar pulsational phenomena like "strange modes".

The `mesalab` pipeline

The pipeline offers several key functionalities

- It collects data on the mass and metallicity ranges of the grid.
- It calculates the necessary bolometric corrections (BCs) by utilizing the MESA Isochrones & Stellar Tracks (MIST) bolometric correction tables (Choi et al., 2016).
- It automatically generates HRDs and Color-Magnitude Diagrams (CMDs) with Gaia passbands.
- Furthermore, `mesalab` provides crucial tools for asteroseismological investigations.

The GYRE module

For blue loop crosser models, the pipeline generates GYRE (Townsend & Teitler, 2013) inlist files with user-specified parameters. In the next step, the pipeline automatically executes GYRE for these pre-filtered models, with the option to run the computations sequentially or in parallel, significantly enhance computational efficiency.

The RSP module

Similar to the GYRE module, the pipeline generates MESA RSP ((Smolec & Moskalik, 2008), (Paxton et al., 2019)) inlist files with user-specified parameters, then executes the MESA RSP module automatically. This process can also be configured to run the computations sequentially or in parallel, with the number of parallel threads also being a user-defined parameter.

Statement of need

The pipeline `mesalab` is a fully automated, Python-based, open-source package built on top of essential packages like Numpy (Harris et al., 2020) and Pandas McKinney (2010). It is designed to simplify the complex and often manual-driven post-processing of stellar evolution grids generated by MESA with a specific focus on blue loop analysis. For the sake of simplicity and for an easy maintability, the package is based on a modular approach.

`mesalab` automates a number of key, repetitive tasks that are essential for the analysis of stellar evolution grids. It automatically identifies various evolutionary phases (and filters out the specific blue loop stage), calculates necessary bolometric corrections, and generates standard diagrams like HRDs and CMDs. Furthermore, `mesalab` significantly enhances the workflow for asteroseismological investigations by automating the generation of inlists and execution of GYRE (Townsend & Teitler, 2013) and MESA RSP ((Smolec & Moskalik, 2008), (Paxton et al., 2019)). This automation, along with the option for parallel execution, dramatically reduces the time and effort required to prepare models for a detailed asteroseismological analysis.

The documentation (https://mesalab.readthedocs.io/en/latest/) of `mesalab` includes a detailed description of the installation, first run, the core functions, along with tutorial notebooks that guide users through its primary functionalities. While these tutorials demonstrate the execution of the workflow, it is important to note that the direct integration with and execution of external tools like GYRE and the MESA RSP modules are not fully reproducible in web-based notebook environments, as they require a local MESA installation.

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References

Choi, J., Dotter, A., Conroy, C., Cantiello, M., Paxton, B., & Johnson, B. D. (2016). Mesa Isochrones and Stellar Tracks (MIST). I. Solar-scaled Models. *The Astrophysical Journal*, *823*(2), 102. https://doi.org/10.3847/0004-637X/823/2/102

Harris, C. R., Millman, K. J., Walt, S. J. van der, Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N. J., Kern, R., Picus, M., Hoyer, S., Kerkwijk, M. H. van, Brett, M., Haldane, A., Río, J. F. del, Wiebe, M., Peterson, P., … Oliphant, T. E. (2020). Array programming with NumPy. *Nature*, *585*(7825), 357–362. https://doi.org/10.1038/s41586-020-2649-2

Jermyn, A. S., Bauer, E. B., Schwab, J., Farmer, R., Ball, W. H., Bellinger, E. P., Dotter, A., Joyce, M., Marchant, P., Mombarg, J. S. G., Wolf, W. M., Sunny Wong, T. L., Cinquegrana, G. C., Farrell, E., Smolec, R., Thoul, A., Cantiello, M., Herwig, F., Toloza, O., … Timmes, F. X. (2023). Modules for Experiments in Stellar Astrophysics (MESA): Time-dependent Convection, Energy Conservation, Automatic Differentiation, and Infrastructure. *The Astrophysical Journal Supplement Series*, *265*(1), 15. https://doi.org/10.3847/1538-4365/acae8d

Joyce, M., Molnár, L., Cinquegrana, G., Karakas, A., Tayar, J., & Tarczay-Nehéz, D. (2024). Stellar Evolution in Real Time. II. R Hydrae and an Open-Source Grid of >3000 Seismic TP-AGB Models Computed with MESA. *The Astrophysical Journal*, *971*(2), 186. https://doi.org/10.3847/1538-4357/ad534a

McKinney, Wes. (2010). Data Structures for Statistical Computing in Python. In Stéfan van

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der Walt & Jarrod Millman (Eds.), *Proceedings of the 9th Python in Science Conference* (pp. 56–61). https://doi.org/10.25080/Majora-92bf1922-00a

Paxton, B., Bildsten, L., Dotter, A., Herwig, F., Lesaffre, P., & Timmes, F. (2011). Modules for Experiments in Stellar Astrophysics (MESA). *The Astrophysical Journal*, *192*(1), 3. https://doi.org/10.1088/0067-0049/192/1/3

Paxton, B., Cantiello, M., Arras, P., Bildsten, L., Brown, E. F., Dotter, A., Mankovich, C., Montgomery, M. H., Stello, D., Timmes, F. X., & Townsend, R. (2013). Modules for Experiments in Stellar Astrophysics (MESA): Planets, Oscillations, Rotation, and Massive Stars. *The Astrophysical Journal Supplement Series*, *208*(1), 4. https://doi.org/10.1088/0067-0049/208/1/4

Paxton, B., Marchant, P., Schwab, J., Bauer, E. B., Bildsten, L., Cantiello, M., Dessart, L., Farmer, R., Hu, H., Langer, N., Townsend, R. H. D., Townsley, D. M., & Timmes, F. X. (2015). Modules for Experiments in Stellar Astrophysics (MESA): Binaries, Pulsations, and Explosions. *The Astrophysical Journal Supplement Series*, *220*(1), 15. https://doi.org/10.1088/0067-0049/220/1/15

Paxton, B., Schwab, J., Bauer, E. B., Bildsten, L., Blinnikov, S., Duffell, P., Farmer, R., Goldberg, J. A., Marchant, P., Sorokina, E., Thoul, A., Townsend, R. H. D., & Timmes, F. X. (2018). Modules for Experiments in Stellar Astrophysics (MESA): Convective Boundaries, Element Diffusion, and Massive Star Explosions. *The Astrophysical Journal Supplement Series*, *234*(2), 34. https://doi.org/10.3847/1538-4365/aaa5a8

Paxton, B., Smolec, R., Schwab, J., Gautschy, A., Bildsten, L., Cantiello, M., Dotter, A., Farmer, R., Goldberg, J. A., Jermyn, A. S., Kanbur, S. M., Marchant, P., Thoul, A., Townsend, R. H. D., Wolf, W. M., Zhang, M., & Timmes, F. X. (2019). Modules for Experiments in Stellar Astrophysics (MESA): Pulsating Variable Stars, Rotation, Convective Boundaries, and Energy Conservation. *The Astrophysical Journal Supplemenet Series*, *243*(1), 10. https://doi.org/10.3847/1538-4365/ab2241

Smolec, R., & Moskalik, P. (2008). Convective Hydrocodes for Radial Stellar Pulsation. Physical and Numerical Formulation. *58*, 193–232. https://doi.org/10.48550/arXiv.0809.1979

team, T. pandas development. (2020). *Pandas-dev/pandas: pandas* (latest). Zenodo. https://doi.org/10.5281/zenodo.3509134

Townsend, R. H. D., & Teitler, S. A. (2013). GYRE: an open-source stellar oscillation code based on a new Magnus Multiple Shooting scheme. *Monthly Notices of the Royal Astronomical Society*, *435*(4), 3406–3418. https://doi.org/10.1093/mnras/stt1533

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